portfolio.

[JULIA WOJACKA]

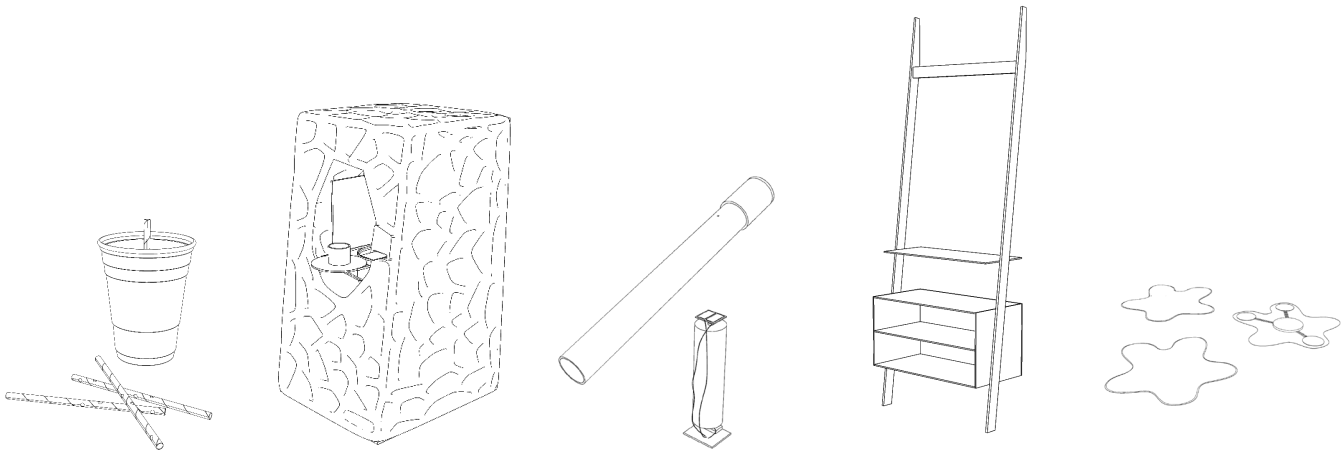
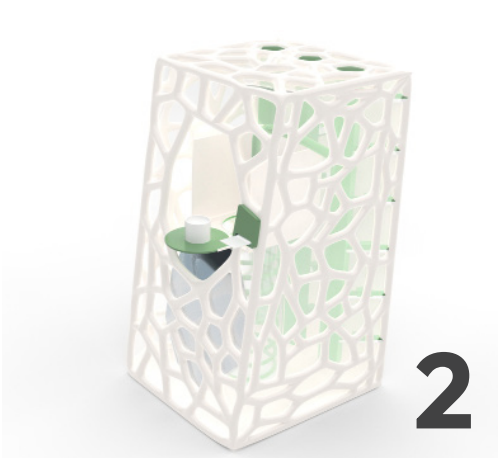


Table of Contents

Inertia



NutriPrint



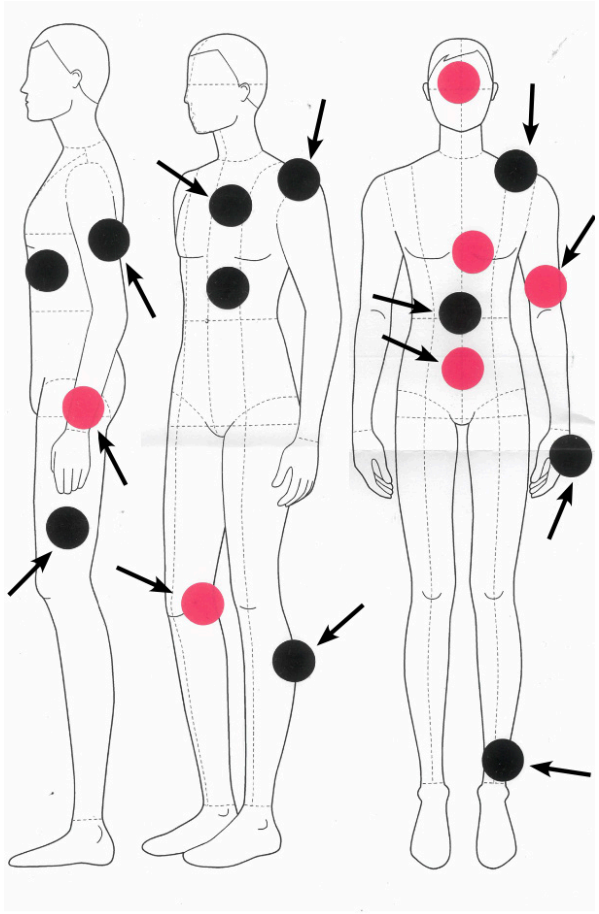
Fluxor

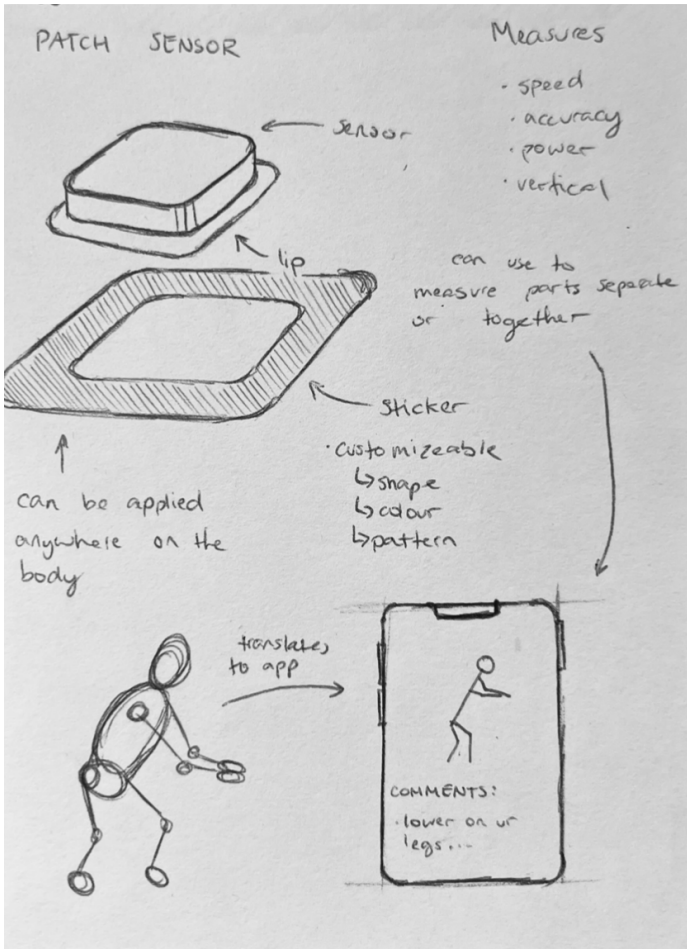


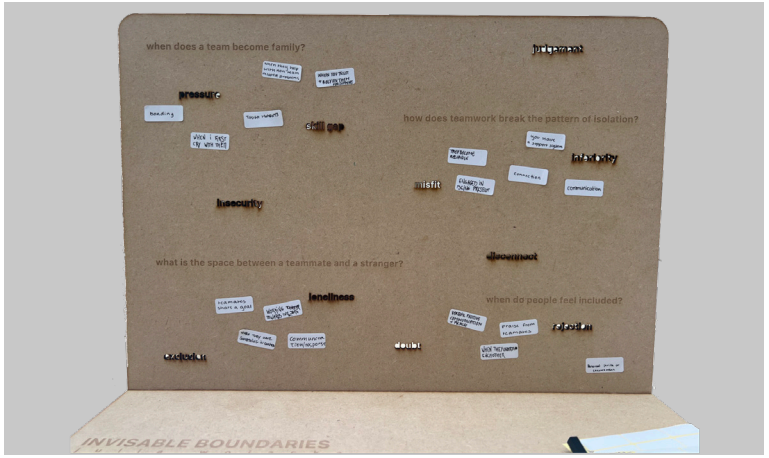
Inertia

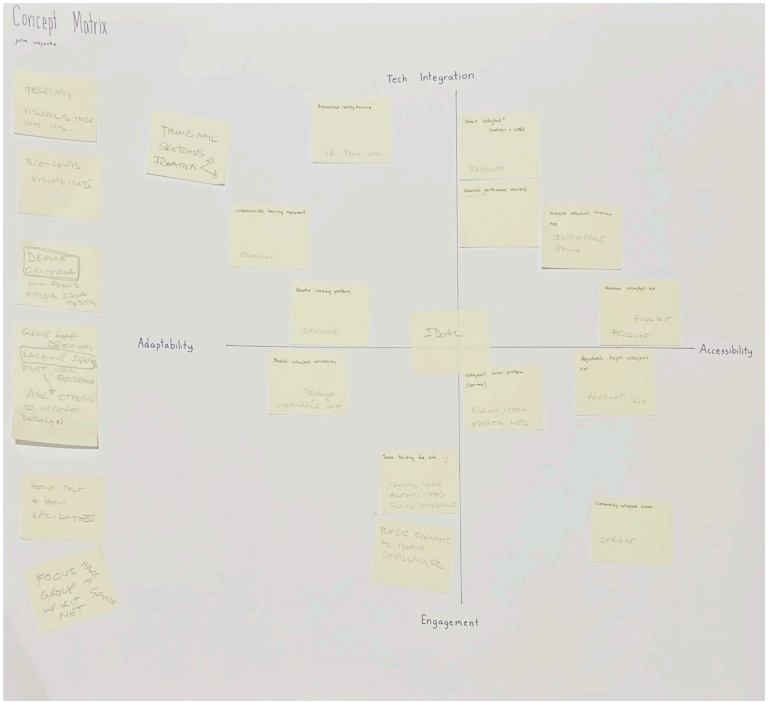
This project was developed in response to a brief focused on team sports, aiming to improve accessibility and inclusion for beginners.

Centered around volleyball, the goal was to reduce barriers to participation—such as fear of judgment and low confidence—by exploring how design can support skill development and foster a stronger sense of community within the sport.







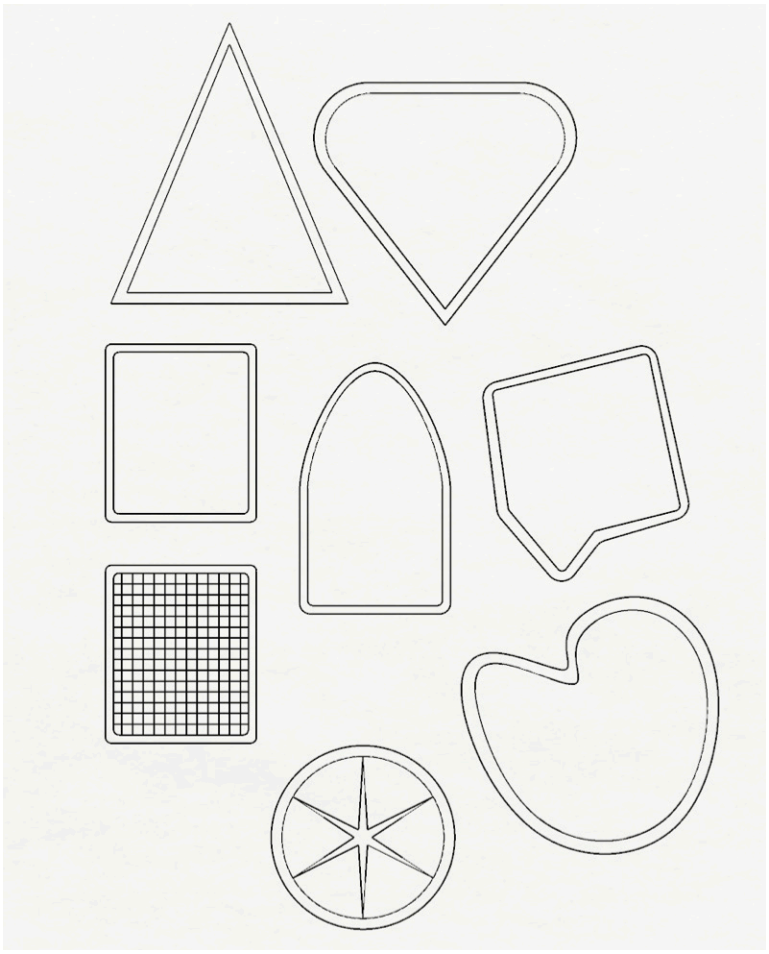


Inertia

Early exploration focused on wearable placement and comfort. I created soft lash-ups and had users test different sizes and positions on the arms, legs, and torso. These trials aimed to identify the most comfortable and unobtrusive areas for movement tracking, laying the groundwork for a design that supports natural play without distraction.



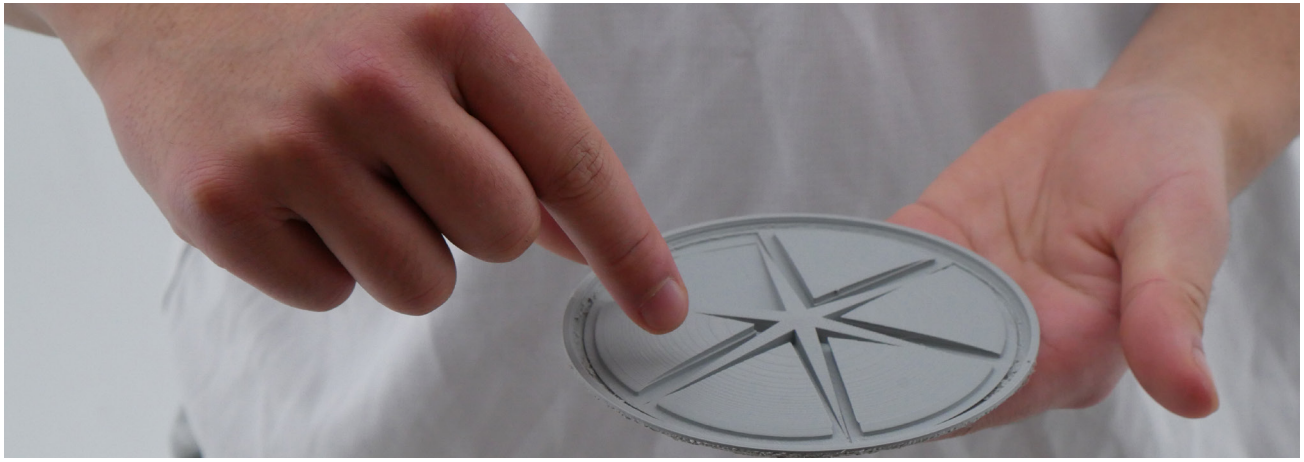




Inertia



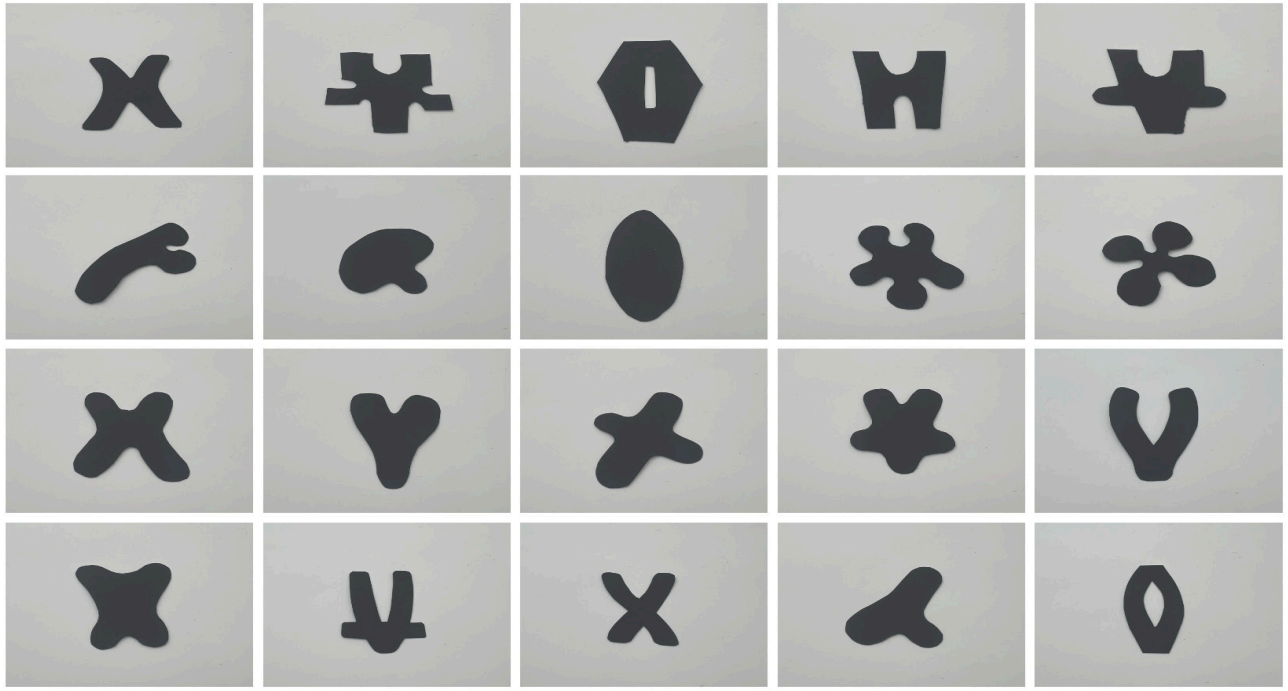




3D-printed prototypes helped explore form, but testing showed that rigid patches were uncomfortable and restrictive. This led to a shift toward flexible, stretchable materials that move naturally with the body for better comfort and performance.



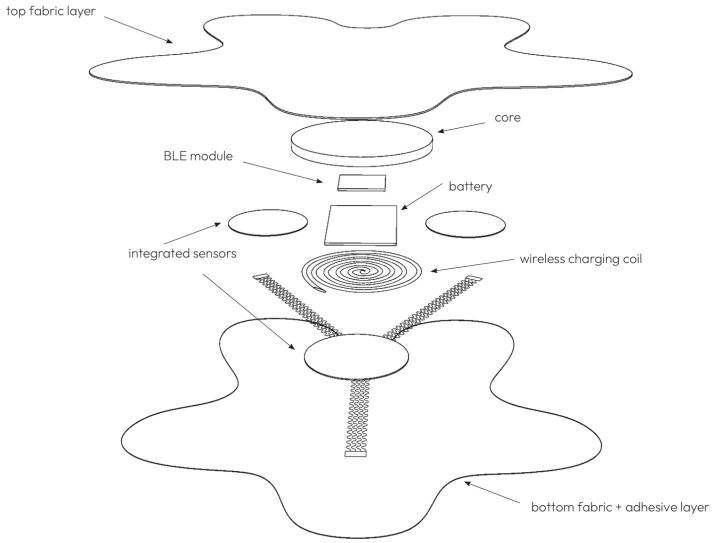
Inertia







User feedback and wearability testing guided the final refinement. I adjusted the shape, scale, and material layering to improve comfort, stability, and movement tracking. The final design balances flexibility with secure adhesion, allowing the patch to stay in place during play while remaining soft, lightweight, and unobtrusive.



Inertia

ideation exploration iteration refinement **final**

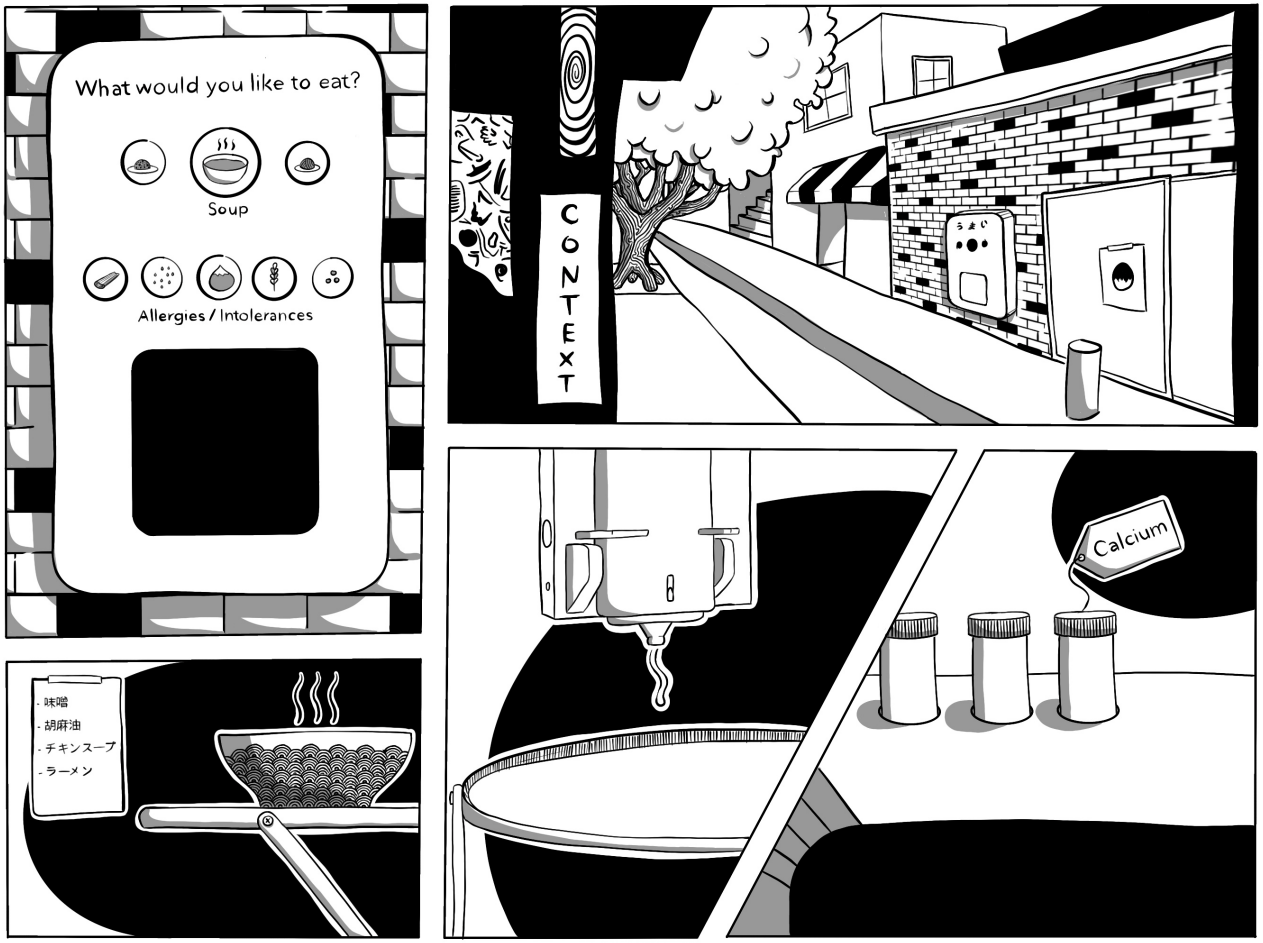


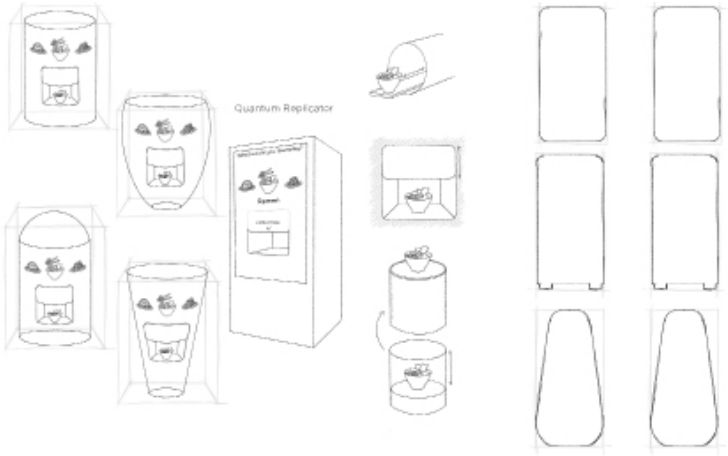


This flexible wearable helps beginners build confidence in volleyball by providing real-time movement feedback. Comfortable, discreet, and intuitive, it supports skill growth and makes the sport more inclusive and approachable.

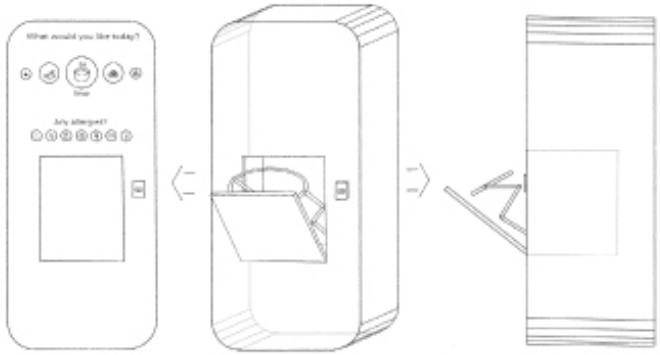
Nutri Print

By 2043, the vegan diet is a symbol of sustainability, driven by the rising impact of the animal industry on the total greenhouse gas emissions over the past 20 years. While plant-based nutrition is slowly becoming the norm, it unfortunately remains financially out of reach for many. However, rapid technological advances, governmental support and social openness are helping drive efforts toward a sustainable future.

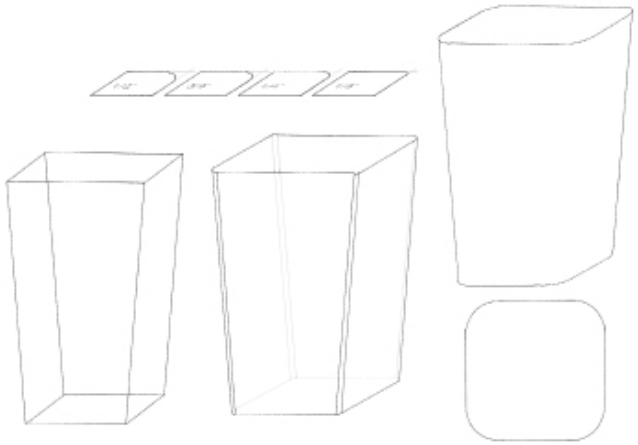


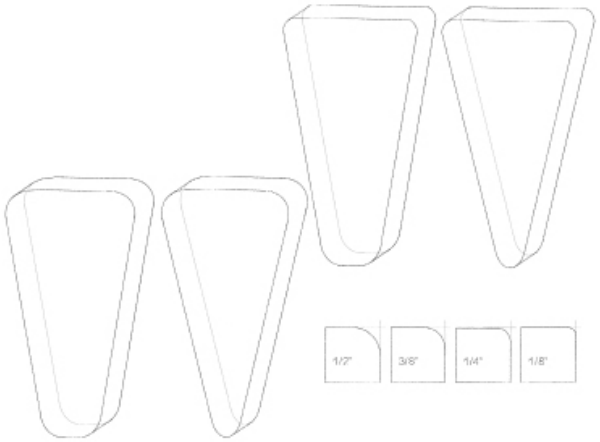


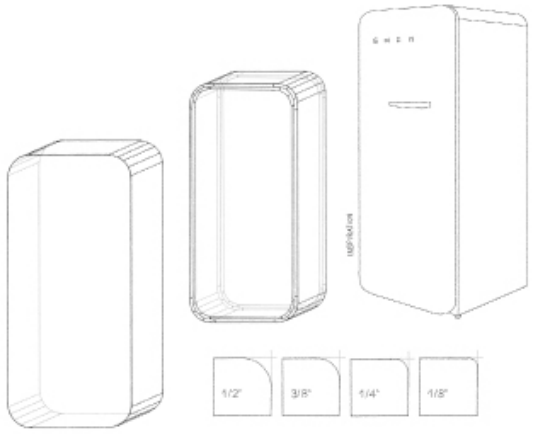
While my original quantum replicator idea was centered around the concept of a K1 civilization, achieving it within 20 years is unrealistic due to current technological, environmental, and socio-political limitations which is why it was just my base in this project.

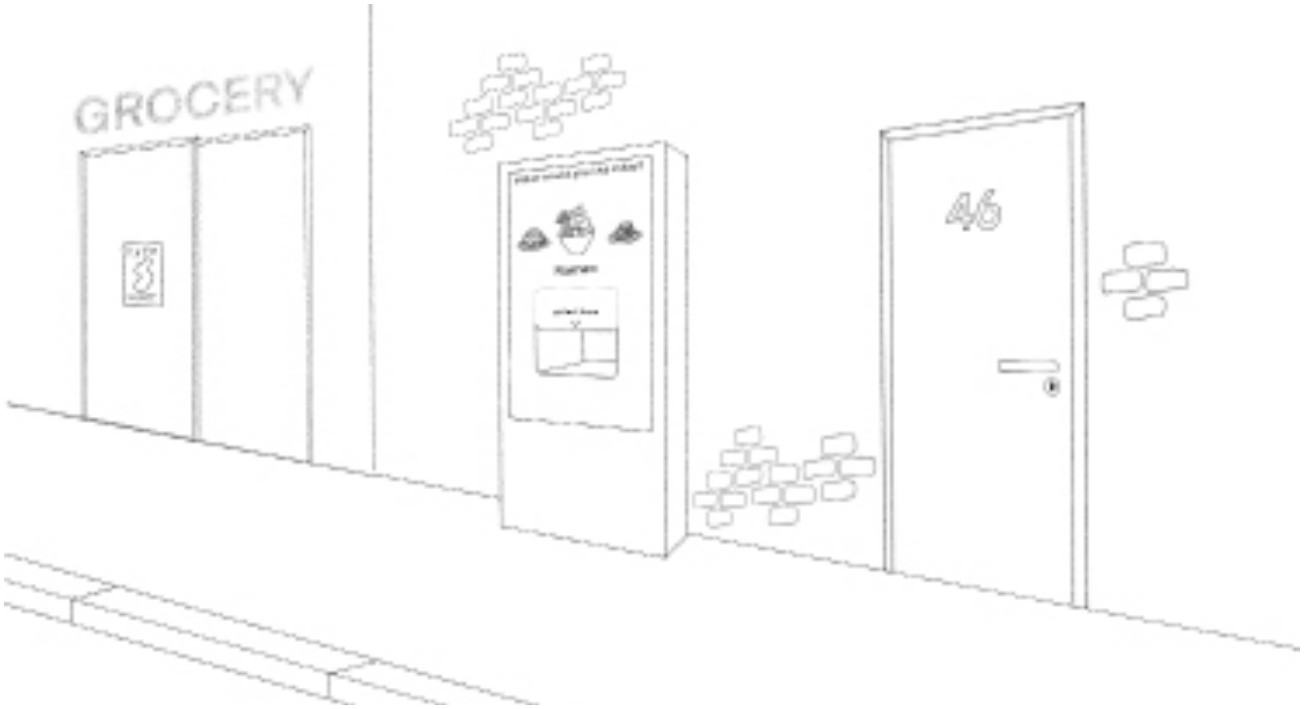


Nutri Print







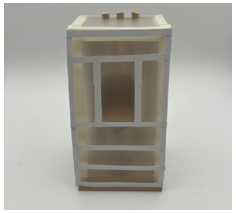


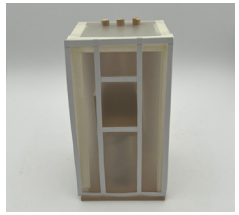
To make the idea more realistic for the next 20 years, I explored molecular gastronomy to create nutrient-rich, smaller-portion meals at lower costs. This concept combines customized nutrition, enhanced flavors, and sustainable sourcing.

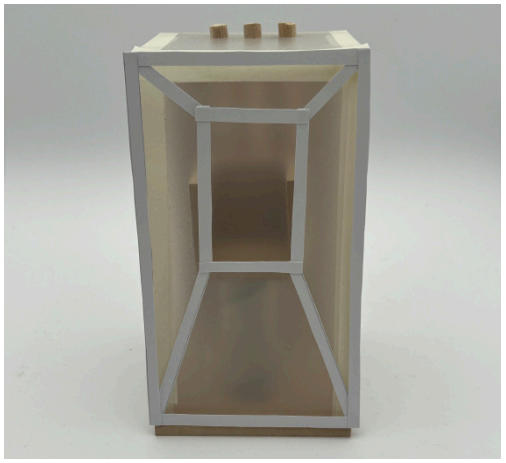
The product is a 3D food dispenser or vending machine that tailors meals to users' dietary needs, delivering all essential nutrients affordably. Subsidized by the government, it will be placed in public spaces like supermarkets, workplaces, and schools, promoting healthier eating and improved community nutrition.

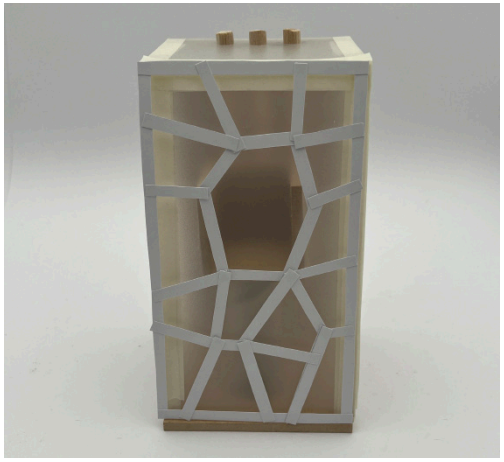
Nutri Print

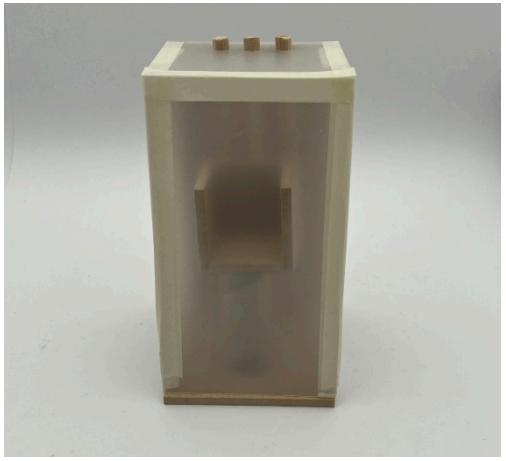
NutriPrint's form was developed using a Voronoi lattice to balance familiarity with innovation. The organic structure draws from natural patterns, creating a form that feels both approachable and new. The lattice was carefully scaled to control surface area, optimize dissolution time, and enhance visual appeal. Overall shape, size, and spacing were. refined to create a product that felt tactile, personal, and efficient in nutrient delivery.

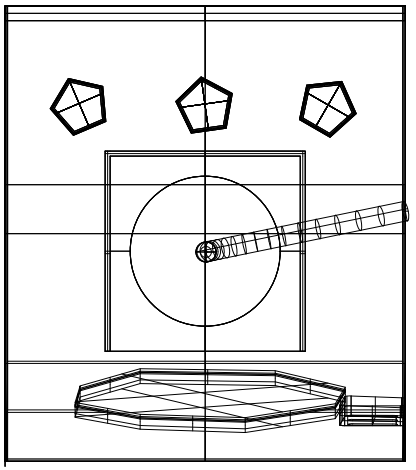


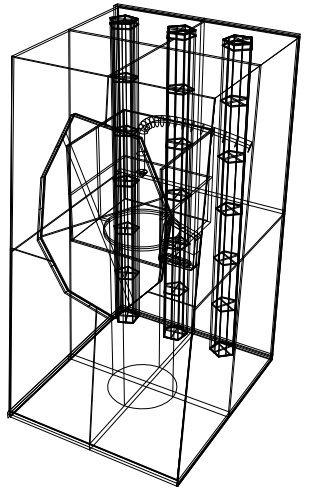


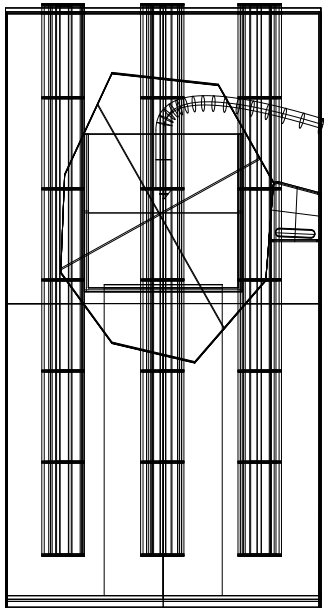


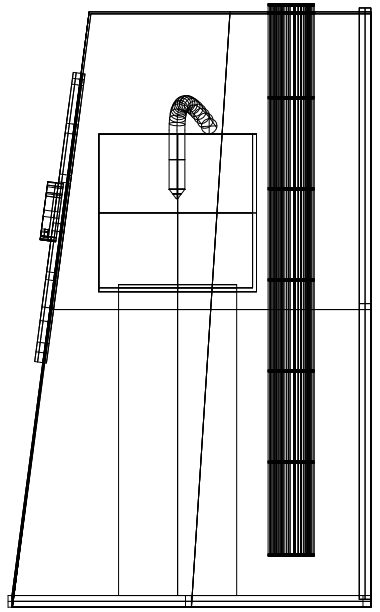






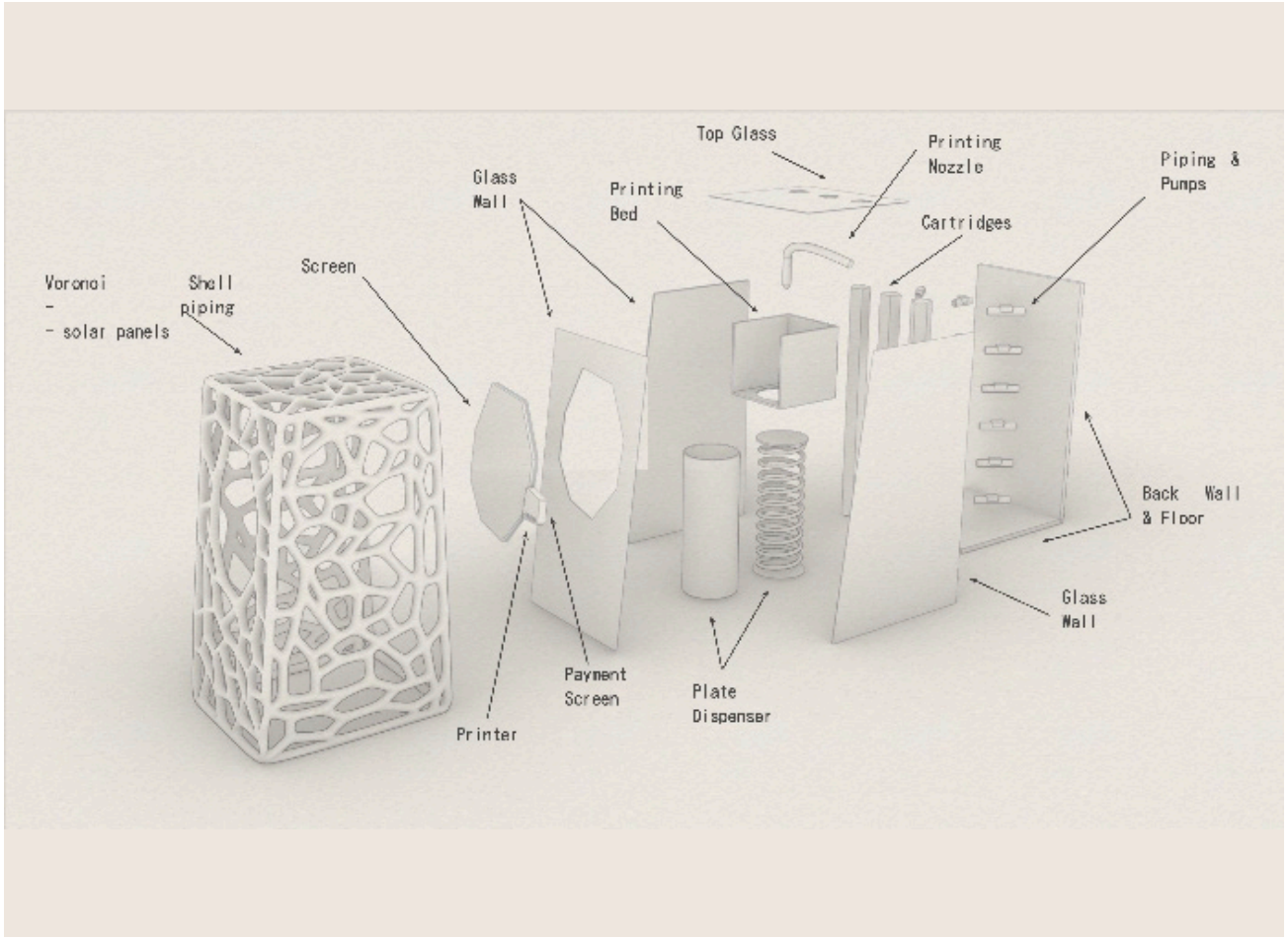


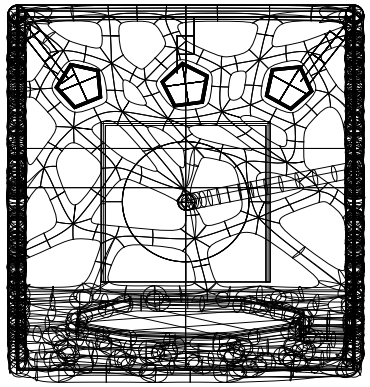


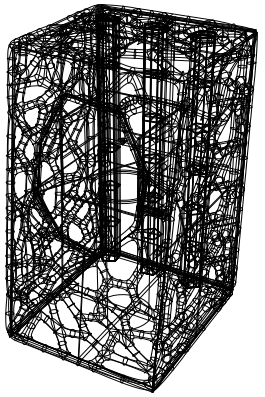


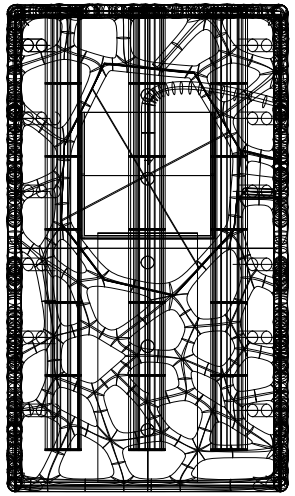
Nutri Print

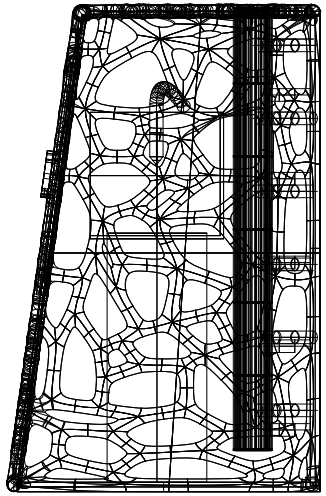
The final NutriPrint design features multiple internal chambers for storing nutrients, each connected to micro-pumps that dispense precise doses. These are combined with edible film through a controlled extrusion system. An interactive touchscreen interface allows users to input personal data or select pre-set health goals, triggering a tailored blend. The compact, modular structure was designed for ease of maintenance and refilling, while the clear casing highlights the internal process—making the technology both functional and engaging.





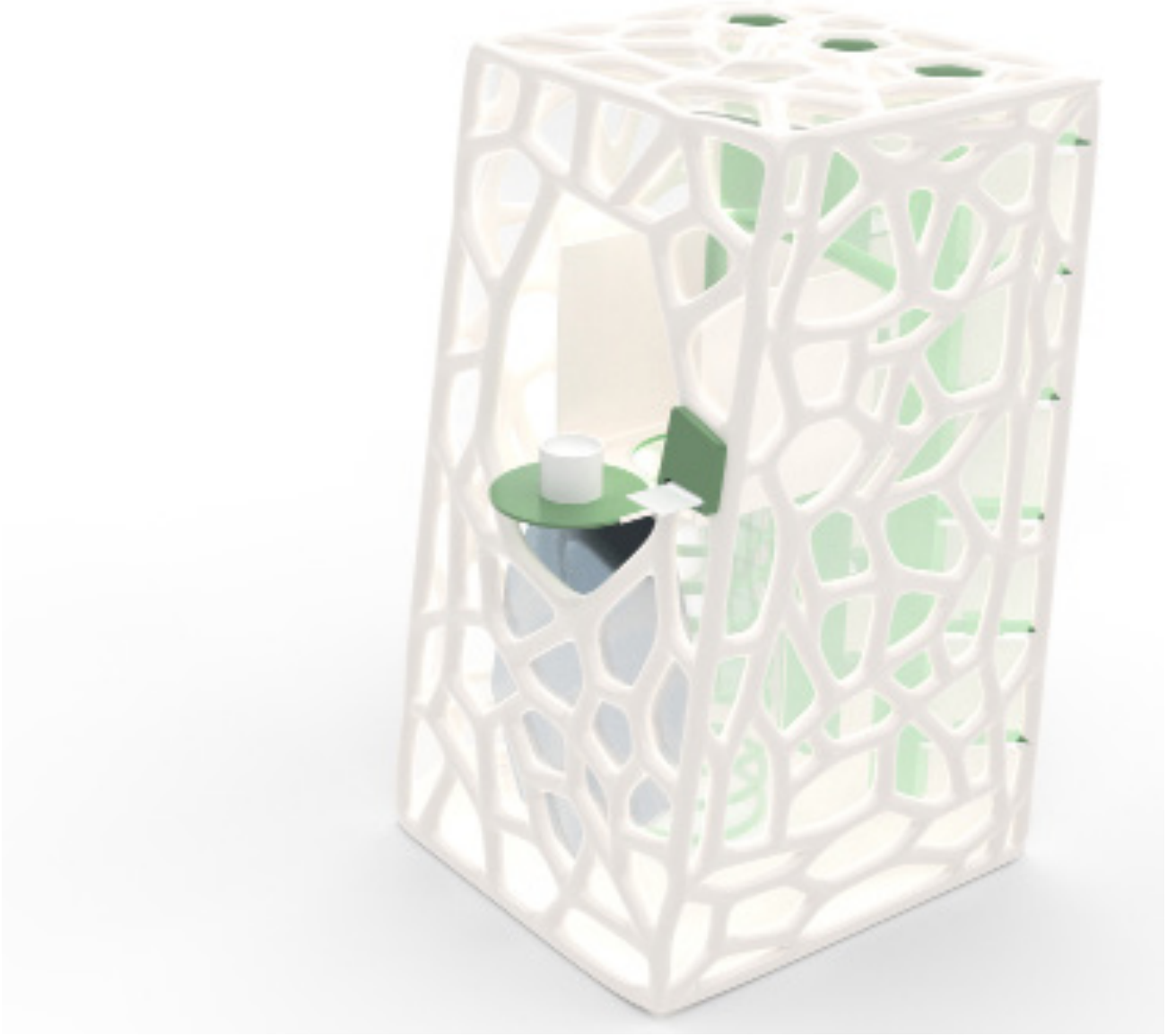






Nutri Print





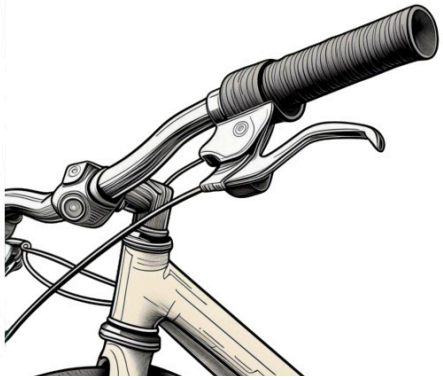
NutriPrint transforms nutrition into a personalized, tactile experience. With smart dosing, intuitive interaction, and a striking Voronoi design, it makes health management efficient, engaging, and human-centered.

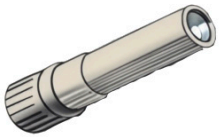
Fluxor



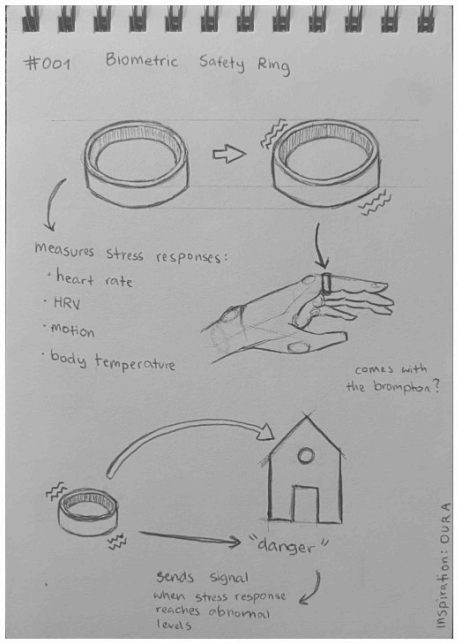


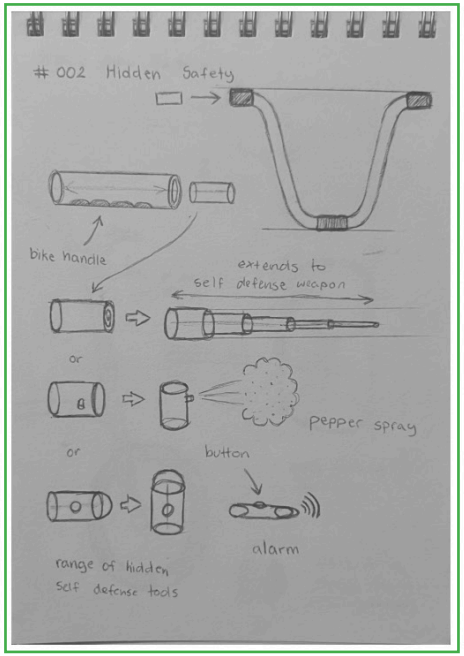


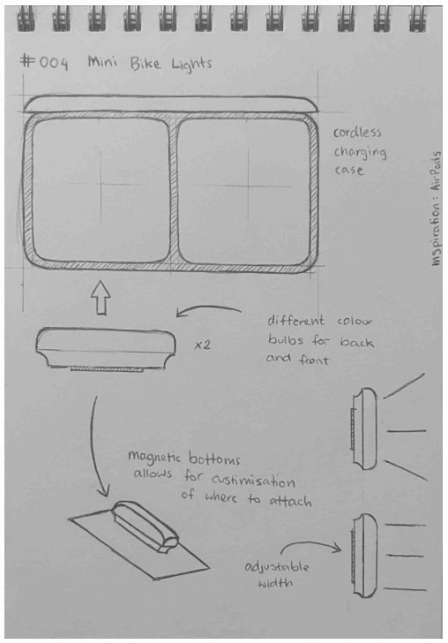


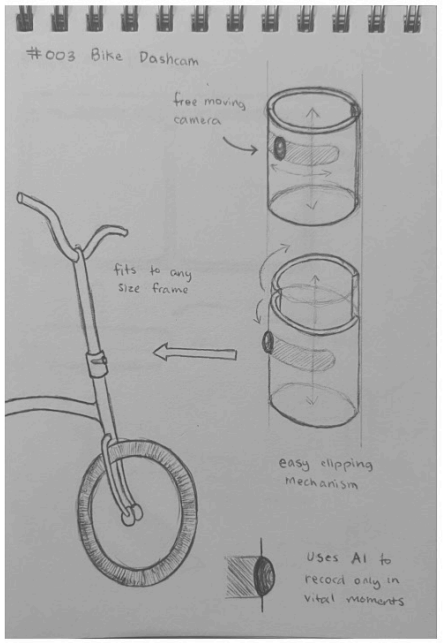


Responding to Brompton's brief, this project explores how design can support women's safety and confidence when cycling at night.









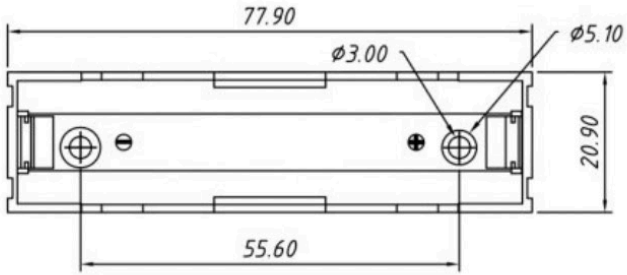
Fluxor

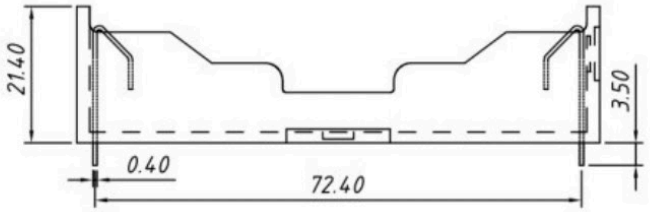


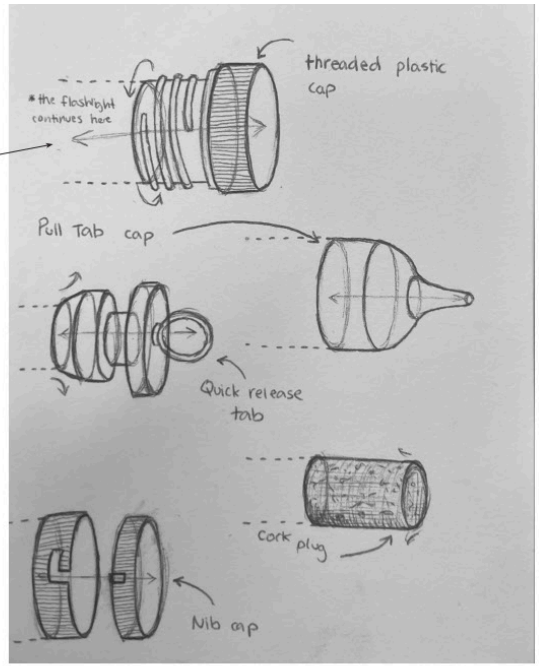
I began by researching Brompton's existing bikes, components, and user base to identify opportunities within the current design. I focused on real user concerns—especially safety, storage, and accessibility—and examined how the bike could be subtly adapted without compromising its core identity. This included studying handlebar structures, lighting options, and mechanisms for easy, instinctive access.



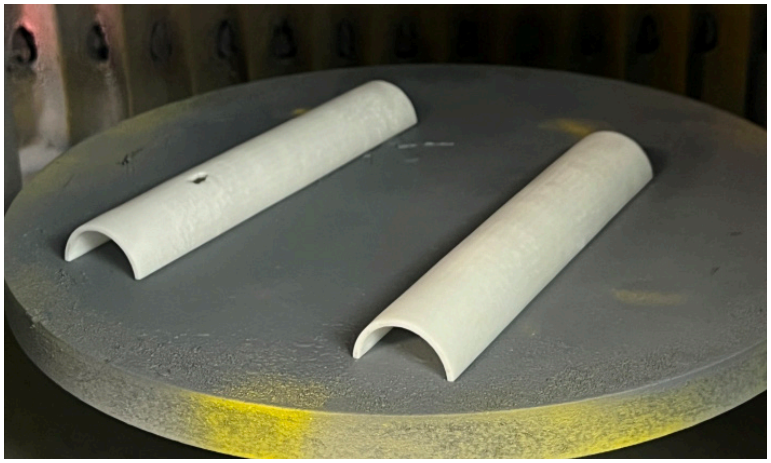


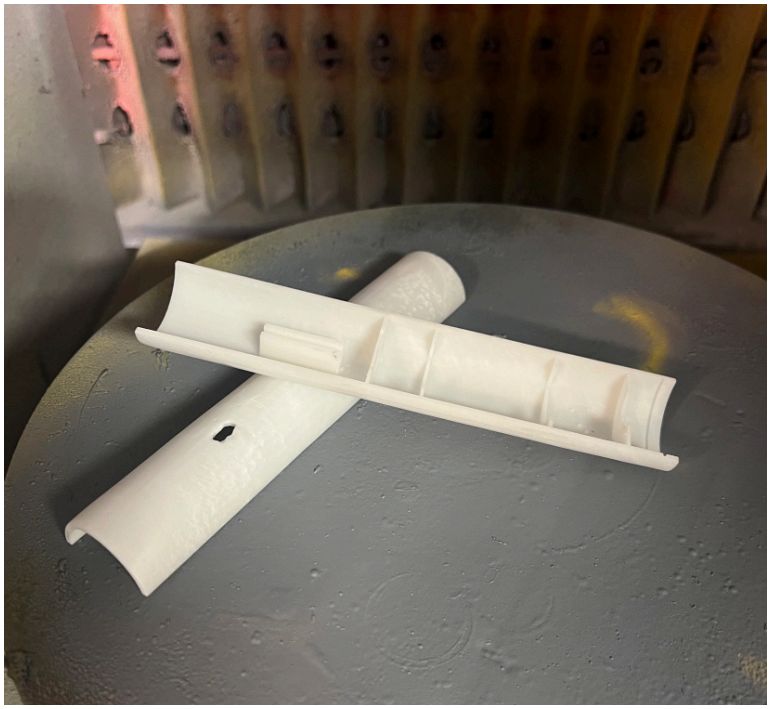




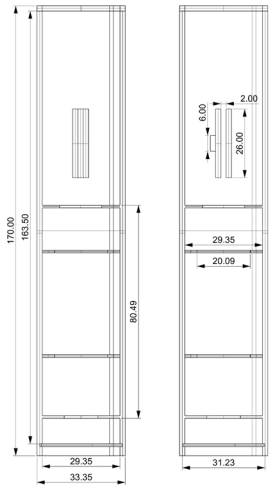


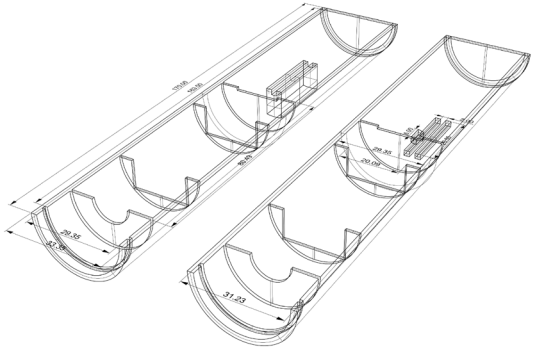
Fluxor



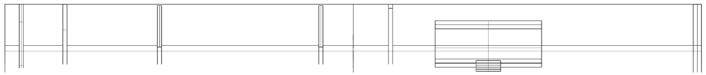


The final model was resin-printed to accurately house internal components, with a cork stopper for secure handlebar integration. It demonstrated the form, function, and ease of access central to the design.



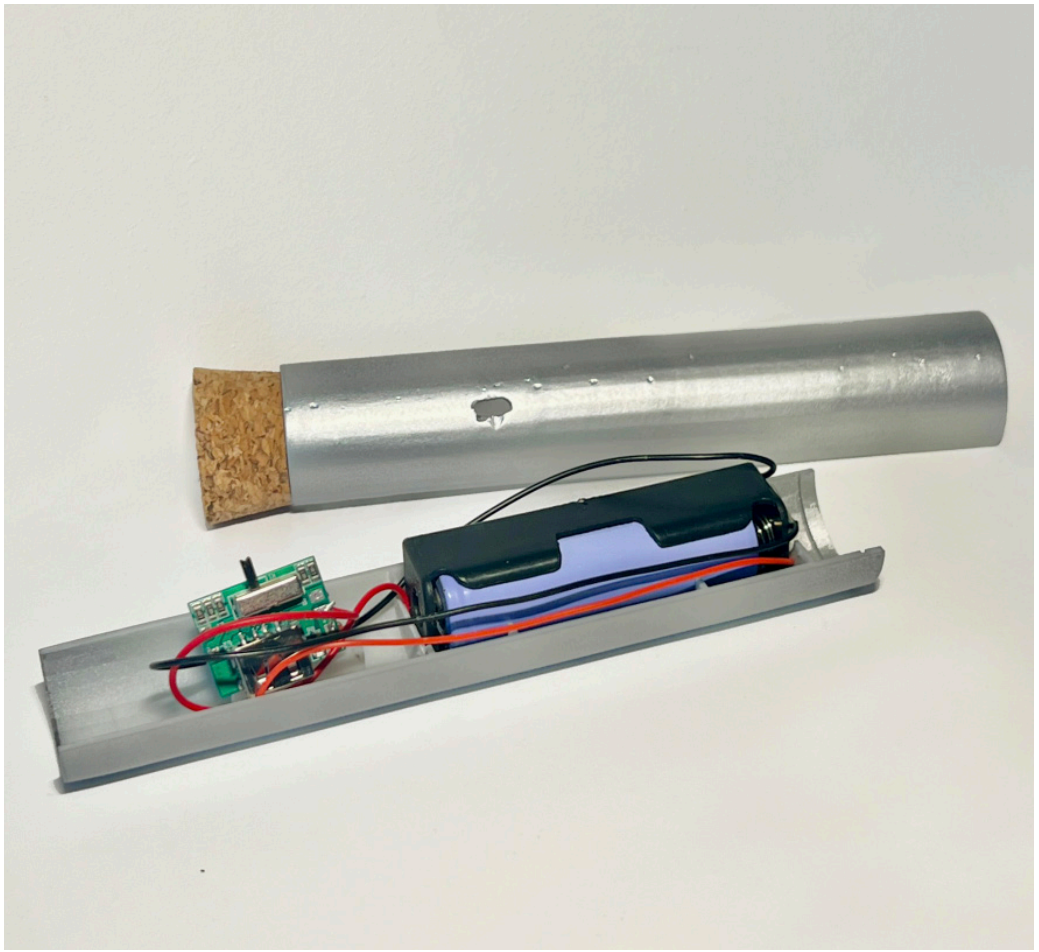


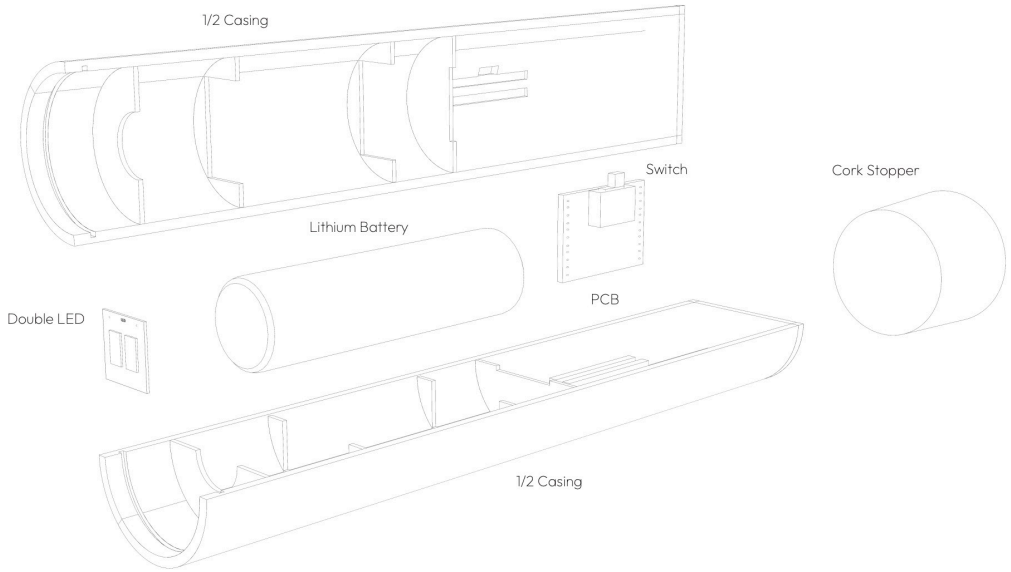


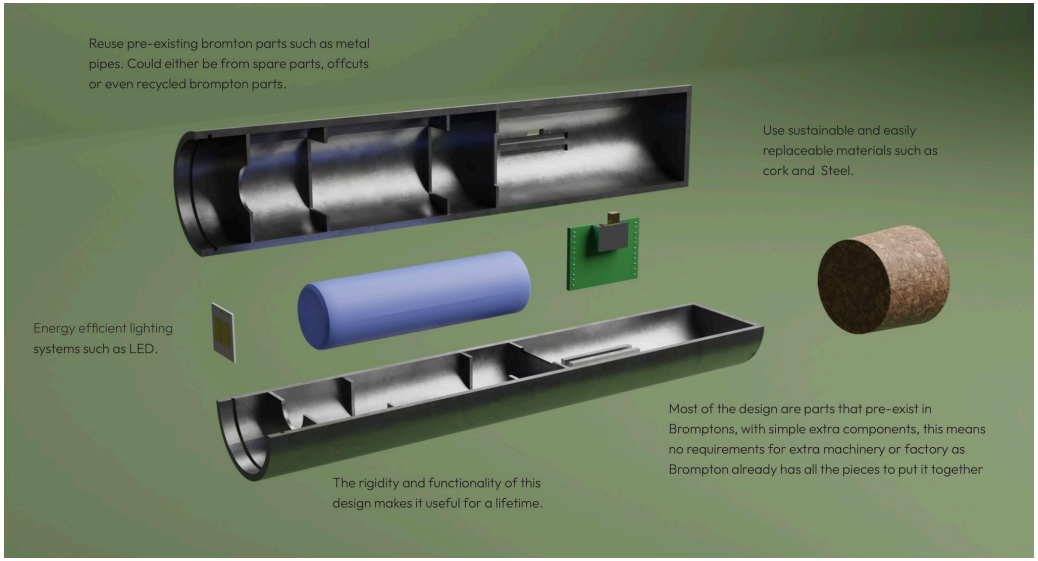


Fluxor

I refined the final model of the casing, ensuring a secure fit for all components and the feasibility of large-scale production. Using Rhino and Blender, I created a realistic prototype, carefully selecting real-world materials that would align with the intended product's final form.







Fluxor



This self-defense flashlight, integrated into a bike handle, offers cyclists quick access to protection. Automatically activating when pulled, it emits a blinding beam to disorient threats, giving vital seconds to escape. Combining practicality with security, it empowers users to navigate the night with confidence.